Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India

(Autonomous College of Affiliated to University of Mumbai)

Mid Semester Examination
September 2024

Max. Marks: 30 | Duration: 1Hr.
Class: T. Y. COMP, AIML, DS | Semester: V
Course Code: CS301 | Date: 23/09/2024
Subject: Theory of Computation

Instructions: (1) All questions are compulsory.
(2) Assume any necessary data but justify the same.
(3) Draw neat and clean diagram.

Q No	Questions	Max Marks	CO	BL
Q1	Let L be a language over the alphabet Σ. It is given that $L^2 = L$. Note that L^2 is the language L concatenated with itself. Which of the following statements must be true? Explain. (A) $L = \Sigma^*$ (B) All strings in L are of even lengths. (C) $L^* = L$ **OR** Let L be a regular language. Prove that $L' = \{w \in L \mid w^R \text{ not in } L\}$ is again regular. where w^R is the reverse of w For example, if w = 011 then w^R = 110.	6	2	3

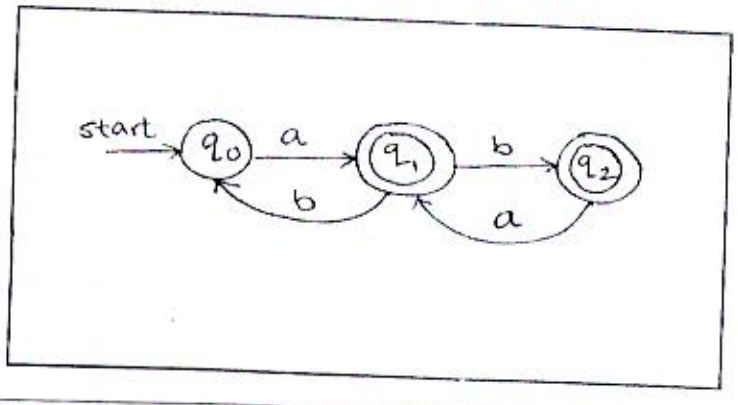
Q2	Write a regular expression for the following language L over the alphabet {a,b}: L consists of all strings which start with ab but do not end with ab	4	1	3
Q3	Let L be the language consisting of all binary strings divisible by 3. For example, 110 belongs to L but 111 does not. Is L regular or not? Explain why or why not.	5	1	3
Q4	Consider the following ϵ-NFA.	10	1	3

	ϵ	a	b	c
→p	{q, r}	{ }	{q}	{r}
q	{ }	{p}	{r}	{p, q}
r*	{ }	{ }	{ }	{ }

(a) Compute the ϵ closure of each state. 1

(b) Give all the strings of length three or less accepted by the automaton 3

(c) Convert the automaton to an NFA without ϵ, and then to a DFA. (Please construct the table and then draw the diagram). 6

Q5	Construct a regular expression for following finite automata. Justify your answer.  	5	1	3